

Chi Lin Kuo (Charlie)

Blockchain & AI & Full-stack Developer

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Summary

- **AI development tooling** — Claude Code and similar agents are my main development partner day to day. I write custom tools and skills for them, which is what shortens the loop from a requirement to a deploy.
- **AI agent development** — I've taken agent systems from zero to one on my own: agent flow design, tool integration, multi-agent orchestration, prompt iteration, run monitoring.
- **Hackathon** — Encode Solana Hackathon, Cypherpunk side track (hosted by Sanctum). Winner, AI finance innovation.
- **Tech Stack**
 - **LLM / Agent** — PydanticAI, LangChain, Google ADK, A2A protocol, MCP, function calling, LLM-as-judge, eval harness
 - **Backend** — Python / FastAPI / Django, Node.js / NestJS, SQLAlchemy 2.0 (async), PostgreSQL / Supabase, Redis, MongoDB
 - **Frontend** — Next.js / React, TypeScript
 - **Other** — Solana / Anchor, Docker, CI/CD, S3

EMPLOYMENT HISTORY

Star Vaults Backend Engineer 03/2026 - 05/2026

Took one complete service module from zero to one (e-commerce platform / internal tooling).

- **Architecture and implementation** — worked through the architecture decisions with one or two senior engineers, then led the implementation, API design, data model and deployment.
- **Backend** — Node.js / NestJS, Python (FastAPI / Django), PostgreSQL, Supabase, Redis, MongoDB.
- **Frontend** — built the matching interfaces in React / Next.js, keeping the data contract consistent across front and back.
- **Process** — boilerplate, unit tests and repetitive code went to Claude Code. Wrote a set of prompt templates and agent skills so that commit messages, code review and tests followed team convention.

Jepun International Backend & Blockchain Engineer 09/2025 - 03/2026

Owned several services and modules in a mid-sized product line (3–5 person team, front and back).

- **Backend** — designed and built RESTful APIs in Python (FastAPI / Django) and Node.js / Express, along with the database schemas behind them (PostgreSQL, MongoDB, Supabase).
- **Third-party integration** — integrated multiple third-party APIs and SaaS services, handling cross-system data sync and error paths.
- **AI / LLM features** — built the LLM features in the product, covering agent flow design and prompt tuning.
- **Frontend** — built the matching interfaces in React / Next.js, keeping the data contract consistent across front and back.
- **DevOps** — automated CI/CD pipelines and containerized deploys.

Projects

Extrovid solo personal project

06/2026 - 07/2026

<https://github.com/KCL1104/extrovid>

An AI-native film production system. It runs script, casting, storyboard, per-shot generation, voice and rough cut end to end, and you can talk to a director agent at any step to change something live.

- **Tech stack** — Python / FastAPI / PydanticAI, SQLAlchemy 2.0 (async) / PostgreSQL, Next.js 16.
- **My work**
 - **Multi-agent orchestration** — 12 PydanticAI agents, with outputs contracted through Pydantic schemas.
 - **Director agent** — carries 8 stateful tools, so the whole pipeline can be driven in plain language.
 - **Pipeline architecture** — a per-stage state machine, so failures resolve to a single stage.
 - **Quality and cost** — LLM-as-judge review with best-of-N selection; a keyframe gate ahead of generation spend.
 - **Testability** — provider factories with `USE MOCK *` flags, so 236 tests run offline.

LorePack Owner / Lead Developer (Personal Project)

02/2026 - 02/2026

<https://github.com/KCL1104/LorePack>

An AI-driven collaborative story world platform. Users build persistent worlds, AI agents generate illustrated chapters, and people collaborate across universes.

- **Tech stack** — Google ADK (multi-agent orchestration framework), A2A protocol, Next.js, FastAPI.
- **My work**
 - Overall architecture and the multi-agent collaboration flow.
 - A2A protocol integration and the inter-agent messaging layer.
 - Frontend.
- **Engineering focus** — orchestration patterns for multi-agent systems, task delegation between agents, and aggregating their results.

DeepFlow Owner / Lead Developer (Personal Project)

01/2026 - 01/2026

<https://github.com/KCL1104/DeepFlow>

A personal project built on the idea of a "smart defence system": context-aware scheduling and dynamic priority filter out notification noise and protect deep work time.

- **Architecture layers**
 - **Frontend** — Next.js (App Router) with shadcn/ui: the "Focus Player" and the queue visualisation.
 - **Backend** — Python FastAPI, Supabase and Redis: priority queue management and multi-platform webhook intake (Slack / Jira / Gmail).
 - **Agent** — Python and LangChain: signal parsing, urgency scoring, context-aware response generation.
- **Technical highlights** — a Redis sorted set (ZSET) gives $O(\log N)$ dynamic priority queueing; LangChain orchestrates the LLM calls.
- **Engineering focus** — the engineering practice behind an AI agent application, from webhook signal intake to agent reasoning flow design to a three-tier service split.

PinTool Backend Engineer (Hackathon-winning / Team Project)

10/2025 - To Present

<https://demo.pintool.fun/>

A Solana-native strategy automation platform. Creators publish on-chain strategies; subscribers pay in USDC on-chain for access.

- **Tech stack** — NestJS, Supabase, Anchor (Solana), TypeScript.
- **My work**
 - Backend architecture, API development, subscription verification logic.
 - Anchor contract integration and on-chain/off-chain state sync.
 - Worked with frontend to get the integration contract right.